# Intelligenza Artificiale

Introduzione agli algoritmi della rivoluzione digitale

UNIVERSITÀ DI PARMA

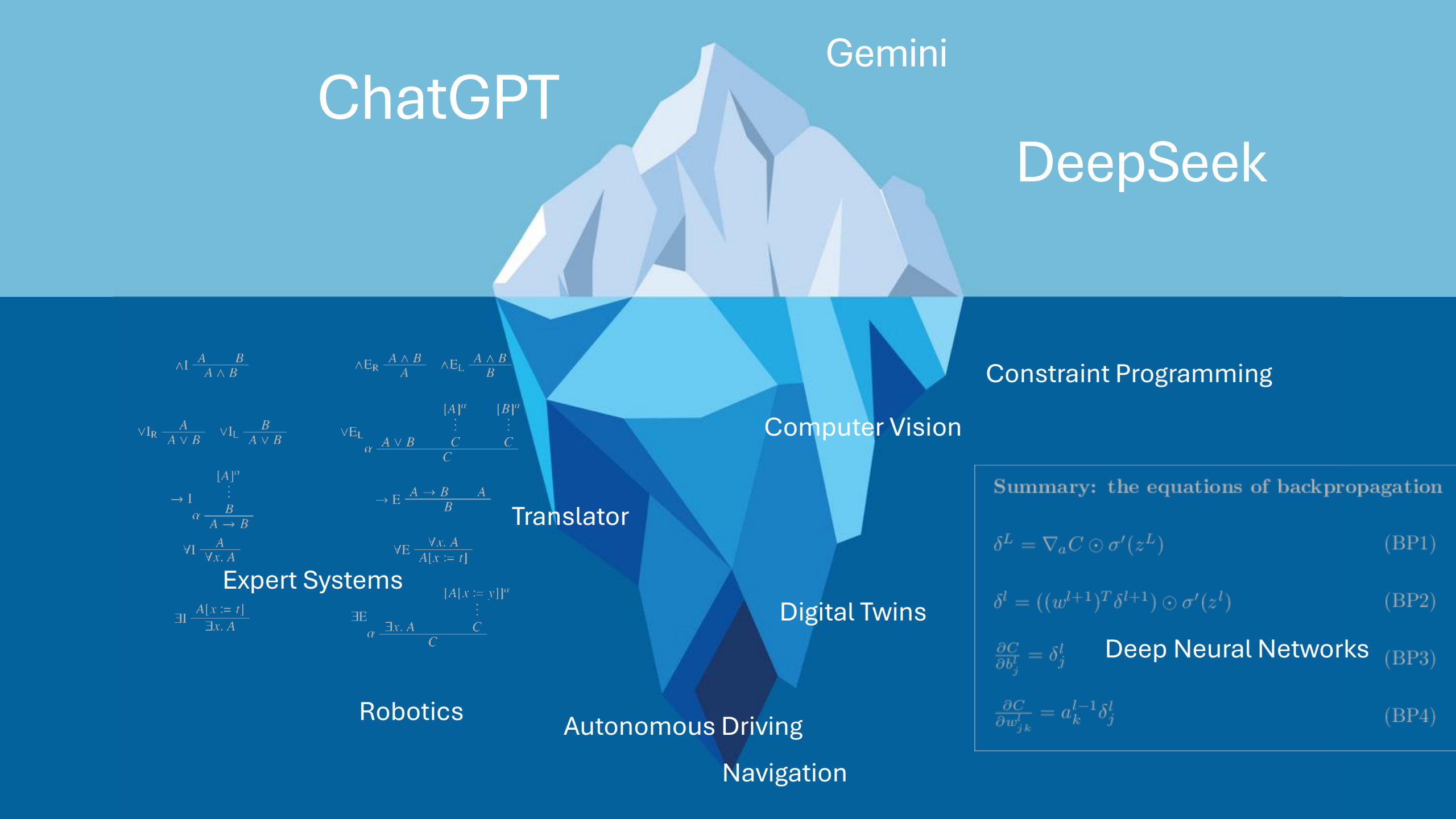
# Intelligenza Artificiale @ UniPR

**Eleonora Iotti**

- PhD in Tecnologie dell'Informazione
- Ricercatrice nell'ambito dell'Intelligenza Artificiale
- Docente di Informatica @ Dipartimento SMFI

# Che cos’è l’Intelligenza Artificiale?

Provate a rispondere voi


ChatGPT
Gemini
DeepSeek
Constraint Programming
Computer Vision
Translator
Expert Systems
Digital Twins
Deep Neural Networks
Robotics
Autonomous Driving
Navigation
Summary: the equations of backpropagation
$\delta^L = \nabla_a C \odot \sigma'(z^L)$ (BP1)
$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$ (BP2)
$\frac{\partial C}{\partial b^l_j} = \delta^l_j$ (BP3)
$\frac{\partial C}{\partial w^l_{jk}} = a^{l-1}_k \delta^l_j$ (BP4)

# Quando è nata l’Intelligenza Artificiale?

Provate a rispondere voi

# Computing Machinery and Intelligence

1950

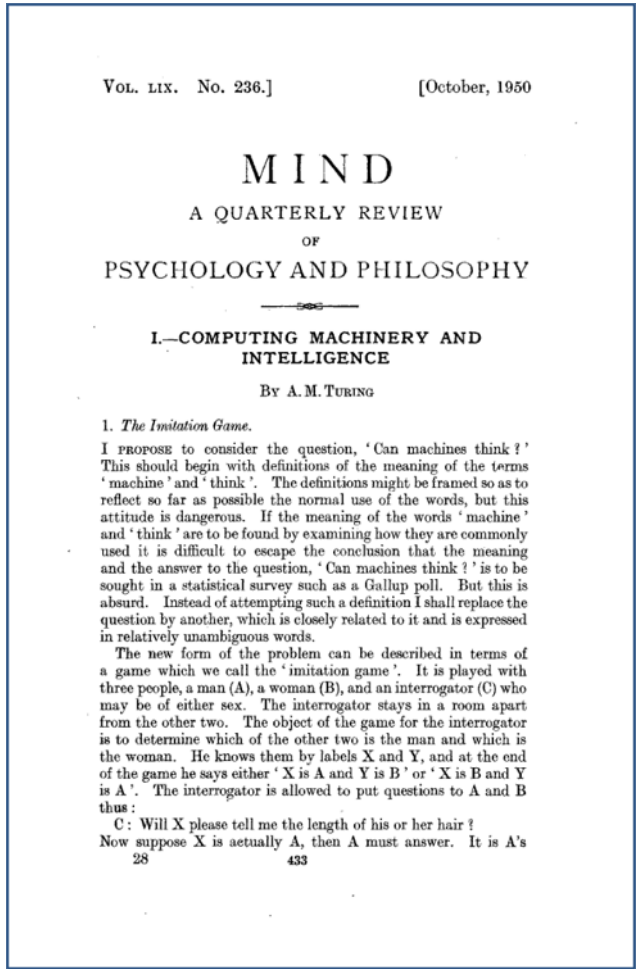
Articolo di Alan Turing

1. The Imitation Game.

I propose to consider the question, ‘Can machines think?’

…

- Le *macchine* possono *pensare*?
  - Se sì, come? Se no, perché?



Vol. lix. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. *The Imitation Game.*

I propose to consider the question, ‘Can machines think?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, ‘Can machines think?’ is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the ‘imitation game’. It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart from the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either ‘X is A and Y is B’ or ‘X is B and Y is A’. The interrogator is allowed to put questions to A and B thus:

C: Will X please tell me the length of his or her hair?

Now suppose X is actually A, then A must answer. It is A's

28 433

# Test di Turing – Imitation Game

- Il computer può dirsi intelligente se in grado di imitare una persona
- Ma cosa caratterizza un *comportamento intelligente*?

Alan Turing

- Propone un test empirico
- Test di Turing è ancora utile per capire di cosa si occupa l'AI

# Di cosa si occupa l’Intelligenza Artificiale?

Provate a rispondere voi

Peter Norvig

Ingegnere e ricercatore a **Google Inc.**

# “Intelligenza Artificiale: un approccio moderno”

Stuart J. Russel

Professore alla **University of California, Berkeley**

| | |
|---|---|
| **Pensare** come una persona | **Pensare** in modo razionale |
| **Agire** come una persona | **Agire** in modo razionale |

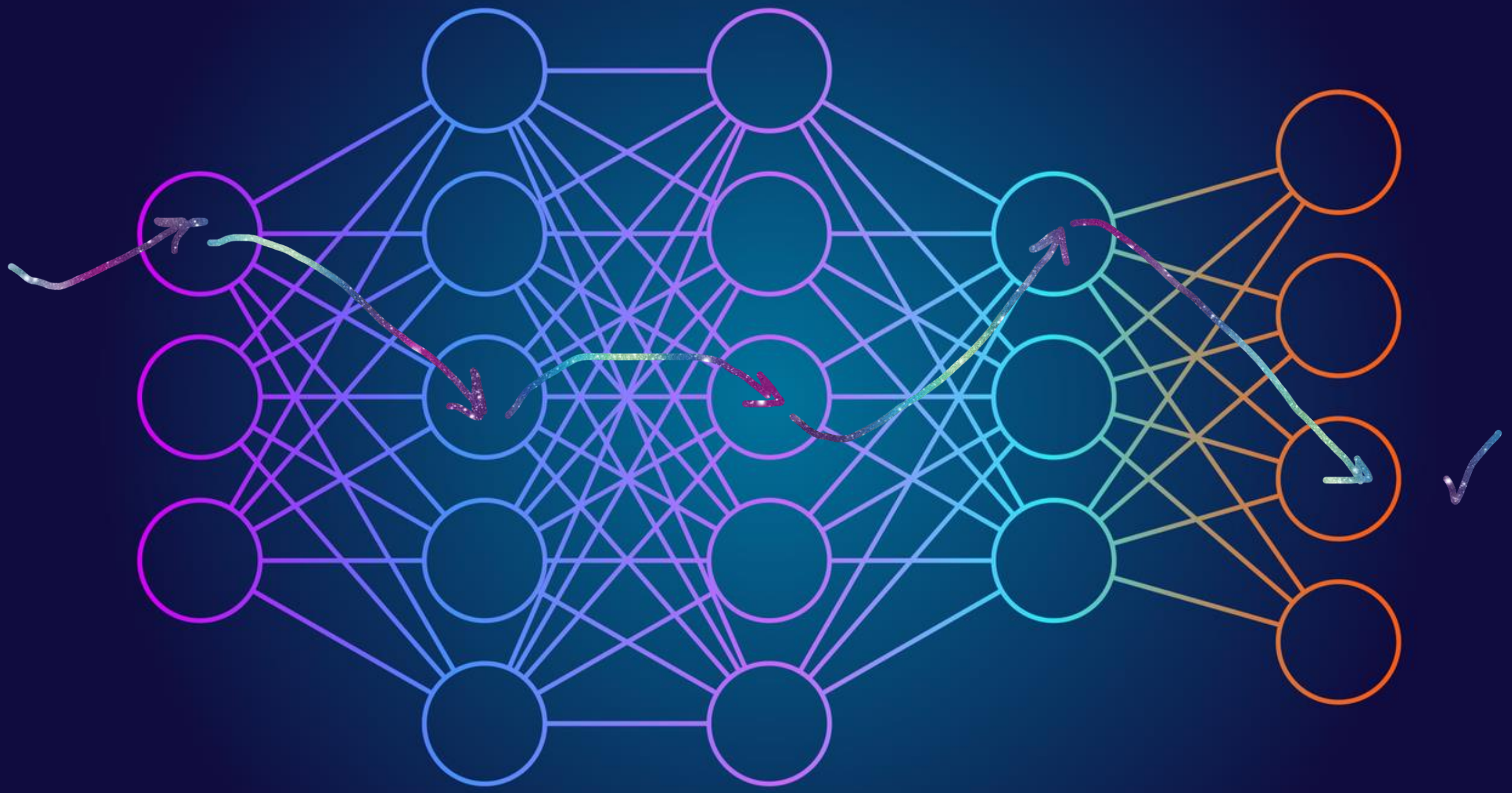
# Il cervello elettronico

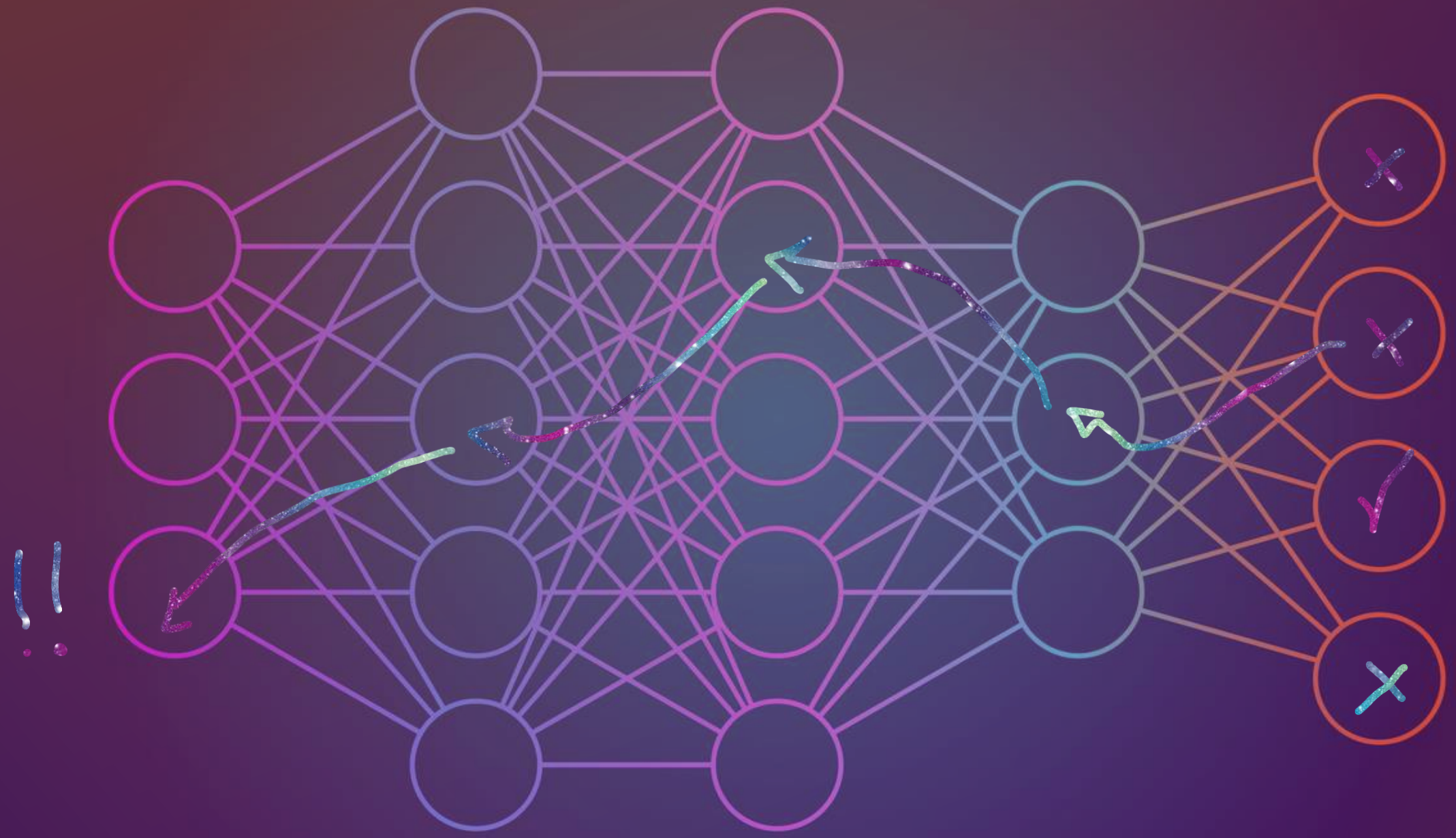
Qual è la tecnologia che sta dietro
all’Intelligenza Artificiale?

# Reti neurali artificiali

## Anni ’70

- Simulare l’organo che genera il pensiero
  - Il cervello e le sue reti neurali
- Obiettivo: creazione di un cervello elettronico
  - Simula il comportamento del cervello umano a livello cellulare
  - Ricostruisce gli equivalenti di neuroni, assoni, soma…
- La rete viene addestrata
  - Processo di *apprendimento*
  - Tramite *esempi* (uso di training set e test set)



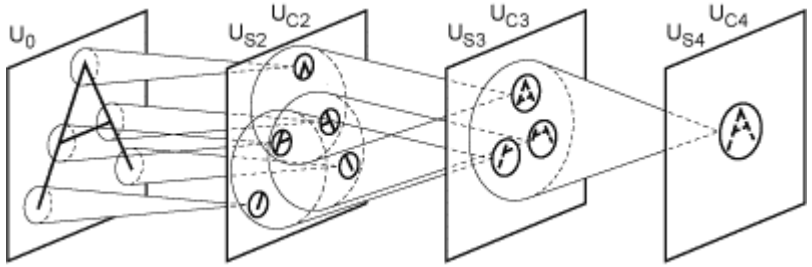


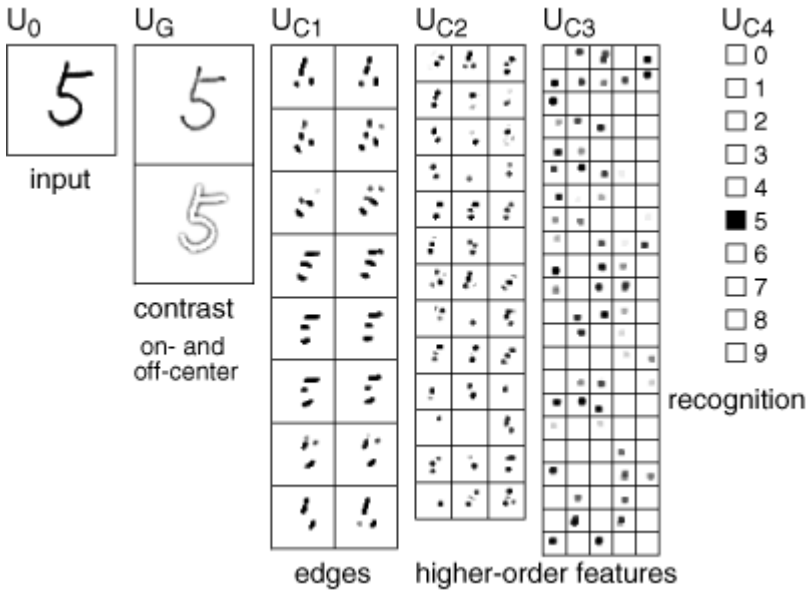
# Reti neurali artificiali

1980

*Neocognitron* (Kunihiko Fukushima)

- Rete neurale addestrata per riconoscere testo manoscritto



# Deep Learning

## Visione artificiale

- Deep neural networks
- ImageNet Large Scale Visual Recognition Challenge

## 1989

***LeNet*** (Yann LeCun)

- Applica l'algoritmo di *backpropagation* a Neocognition

## 2019

- **Premio Turing** a Yann LeCun, Yoshua Bengio, Geoffrey Hinton
    - Per il loro lavoro sul Deep Learning

# Deep Learning

2012

*AlexNet*

(Alex Krizhevsky)

2014

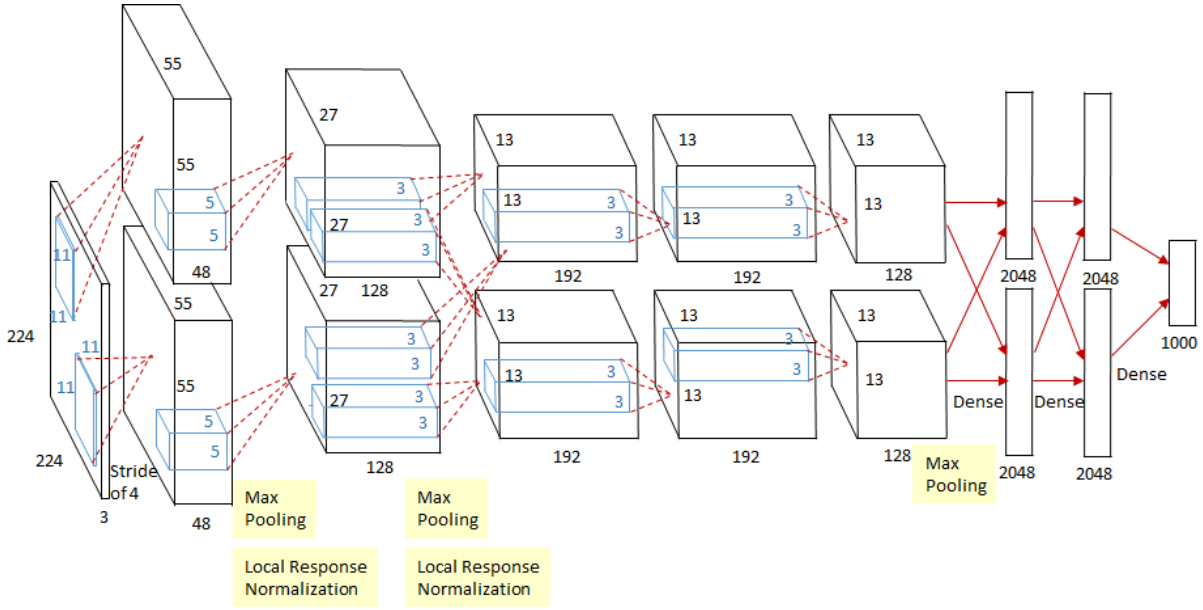
*VGG16, VGG19*

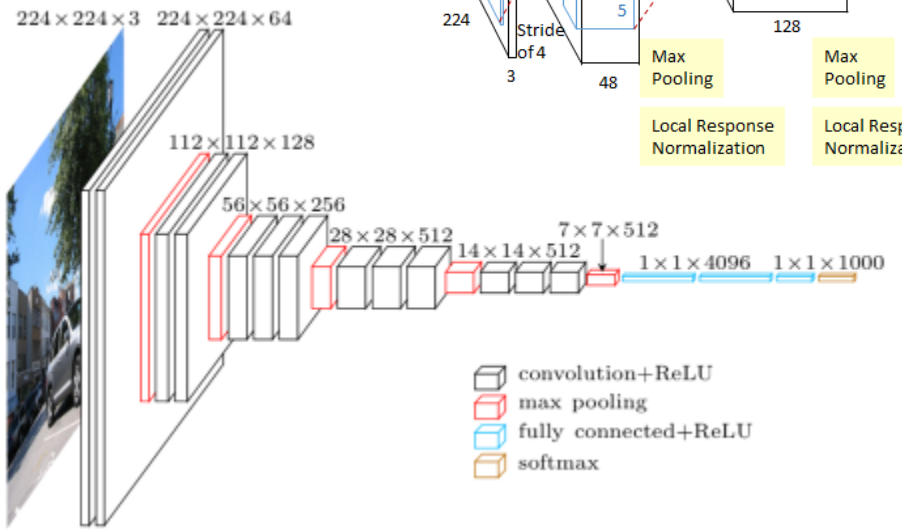
(Karen Simonyan, Andrew Zisserman)

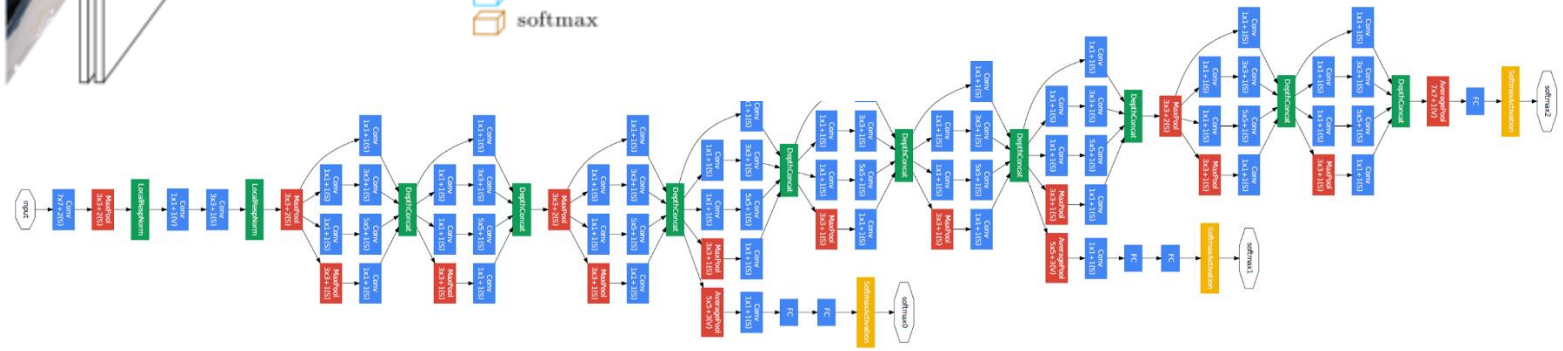
2015

*Inception*

(Google Inc.)



# Dal riconoscimento… alla generazione

2013

*Variational Autoencoders*

(Diederik P. Kingma e Max Welling)

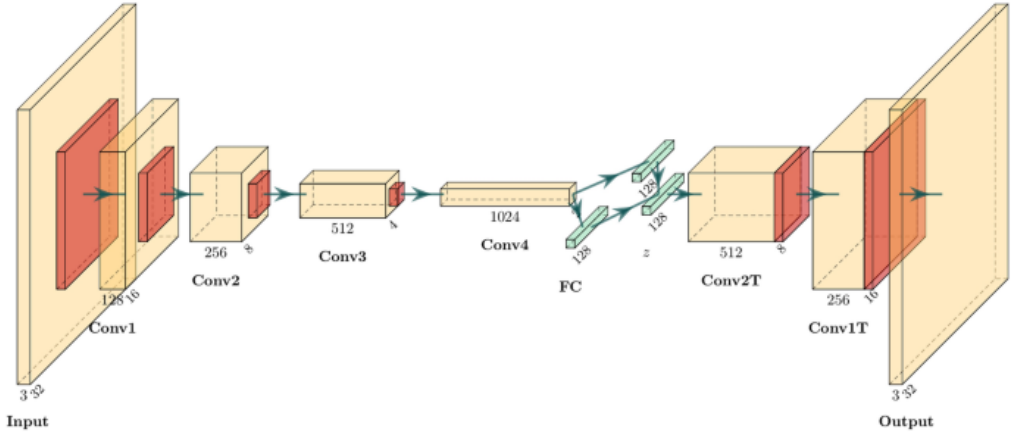
2014

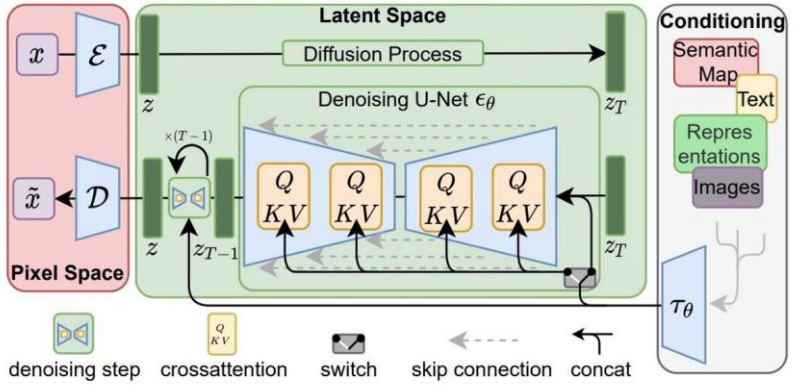
*Reti generative avversarie – GAN*

(Ian Goodfellow)

2015

*Diffusion models*



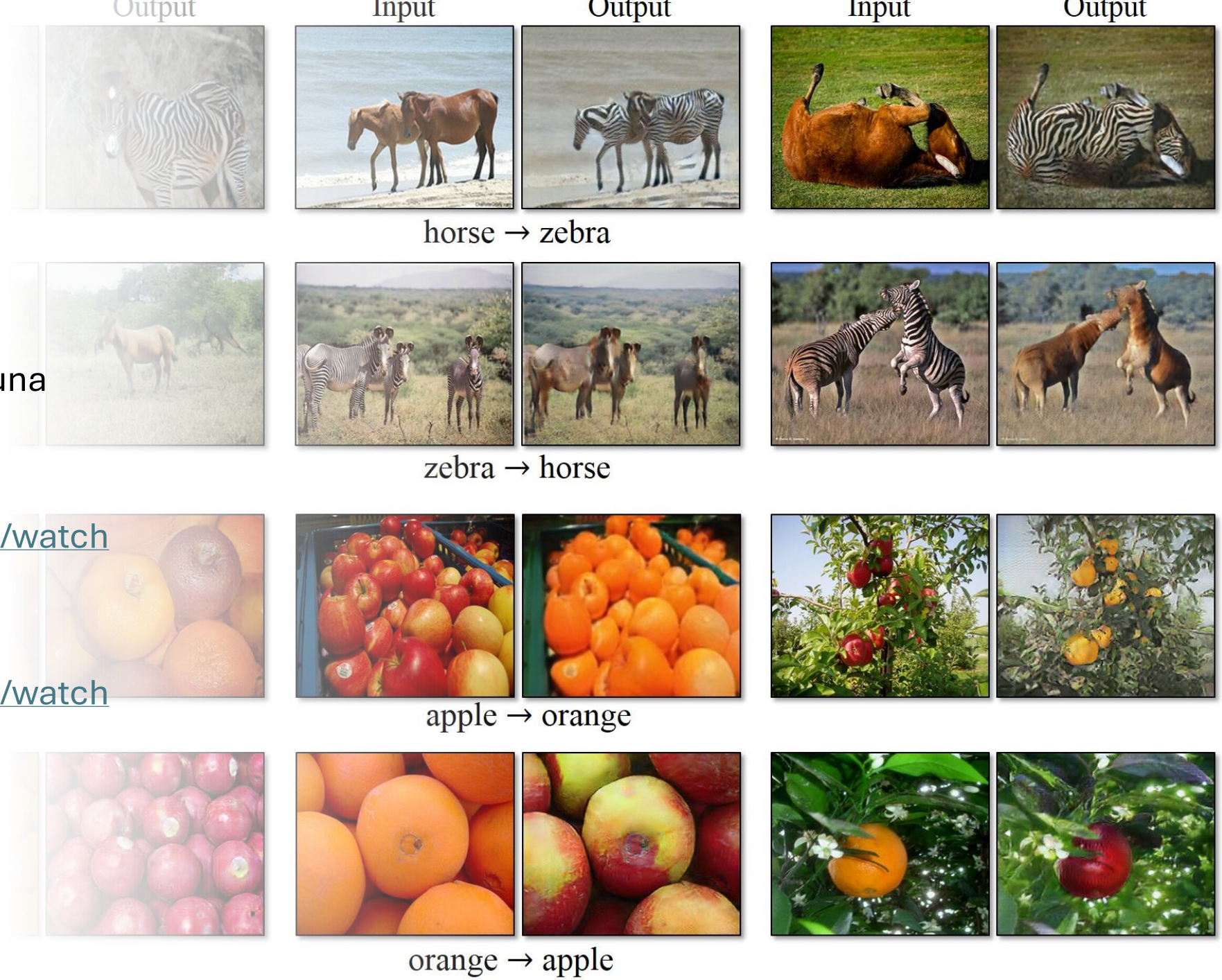



# Cycle GAN

Un cavallo può diventare una zebra?

https://www.youtube.com/watch?v=9reHvktowLY&t=3s

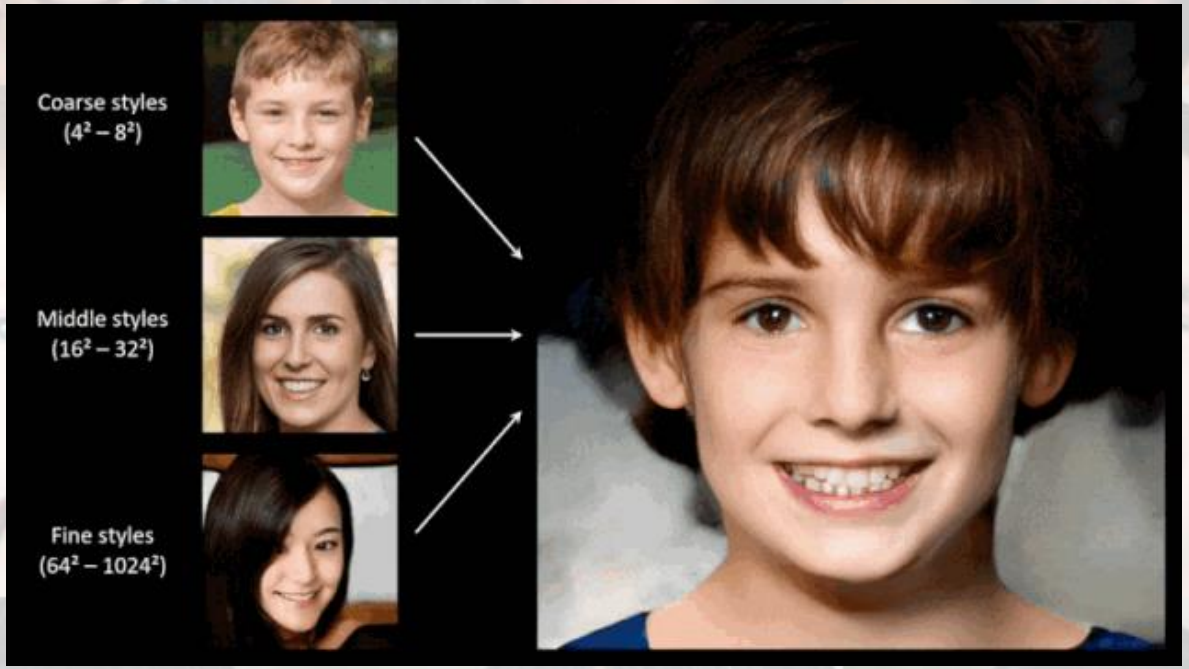
https://www.youtube.com/watch?v=05zzhkyofLE&t=8s



# Style GAN

2018

- Generazione di immagini di volti (fake) in alta definizione
  - NVIDIA Corporation



# Creare immagini artistiche con l'AI

2022

*DALL-E*

(OpenAI)

https://labs.openai.com/

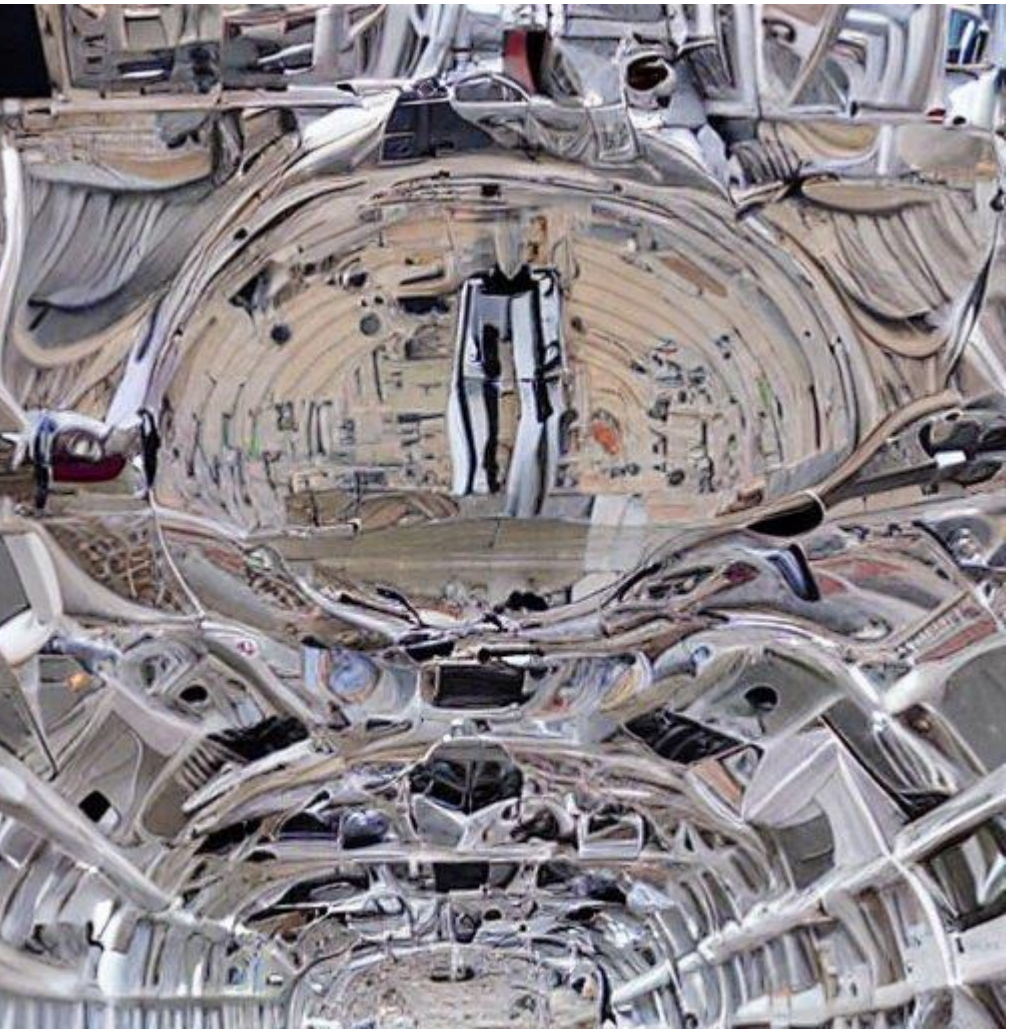
*Midjourney*

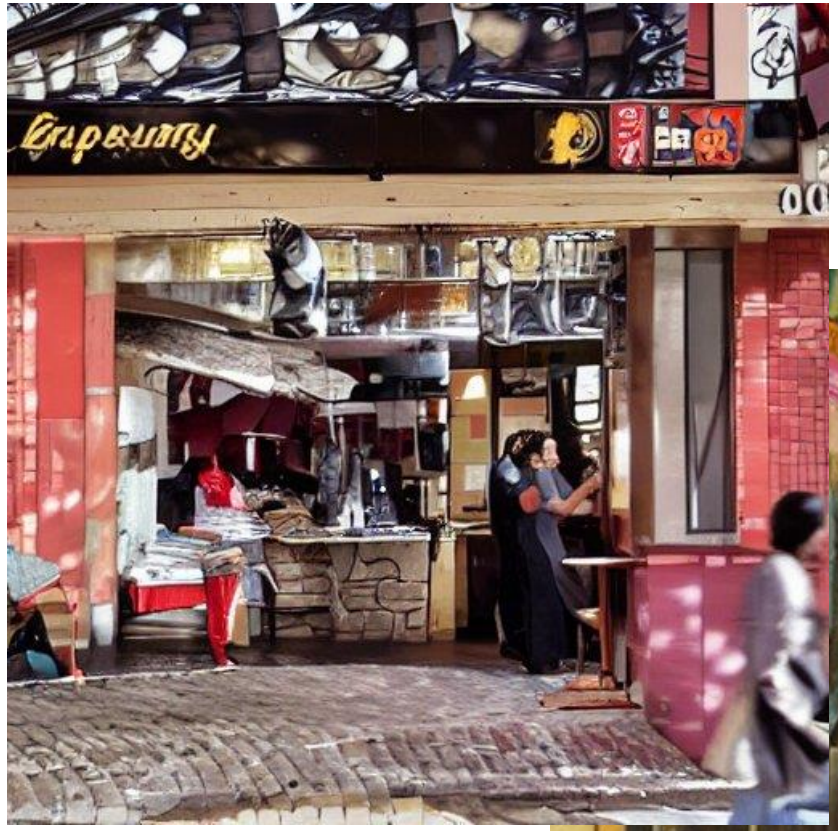
https://www.midjourney.com/home/?callbackUrl=%2Fapp%2F

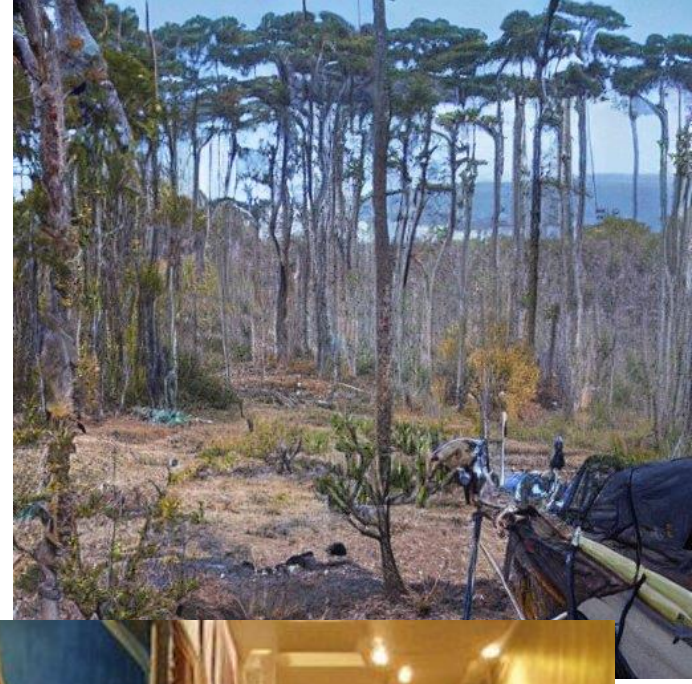
*StableDiffusion*

https://stablediffusionweb.com/#demo

# Imitano l’artista, ma…







# Perché succede?

Provate a rispondere voi

# Sistemi esperti

- *Logica*: studio dei ragionamenti corretti
- Il ragionamento *razionale* è solo quello della Logica
  - Ritengo che se è vero A allora lo è anche B
  - Appena ritengo vero A devo ritenere vero B

- La Logica consente di fare delle *deduzioni*

$$\wedge\mathrm{I}\ \frac{A \qquad B}{A \wedge B} \qquad \wedge\mathrm{E}_{\mathrm{R}}\ \frac{A \wedge B}{A} \qquad \wedge\mathrm{E}_{\mathrm{L}}\ \frac{A \wedge B}{B}$$

$$\vee\mathrm{I}_{\mathrm{R}}\ \frac{A}{A \vee B} \qquad \vee\mathrm{I}_{\mathrm{L}}\ \frac{B}{A \vee B} \qquad \vee\mathrm{E}_{\alpha}\ \frac{A \vee B \qquad \begin{array}{c}[A]^{\alpha}\\ \vdots \\ C\end{array} \qquad \begin{array}{c}[B]^{\alpha}\\ \vdots \\ C\end{array}}{C}$$

$$\rightarrow\mathrm{I}_{\alpha}\ \frac{\begin{array}{c}[A]^{\alpha}\\ \vdots \\ B\end{array}}{A \rightarrow B} \qquad \rightarrow\mathrm{E}\ \frac{A \rightarrow B \qquad A}{B}$$

$$\forall\mathrm{I}\ \frac{A}{\forall x.\, A} \qquad \forall\mathrm{E}\ \frac{\forall x.\, A}{A[x := t]}$$

$$\exists\mathrm{I}\ \frac{A[x := t]}{\exists x.\, A} \qquad \exists\mathrm{E}_{\alpha}\ \frac{\exists x.\, A \qquad \begin{array}{c}[A[x := y]]^{\alpha}\\ \vdots \\ C\end{array}}{C}$$

# Programmazione a vincoli

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 5 | 8 | | 1 | 2 | | | | 9 |
| | | 2 | | 4 | | 5 | 3 | |
| 4 | | 7 | | | | 1 | | 2 |
| 6 | 2 | | 8 | | | | | 5 |
| | | | 2 | | | | 9 | 1 |
| | 5 | | | | 4 | | | |
| 8 | 6 | | | 7 | 2 | 9 | | 3 |
| | | | 9 | 5 | 1 | 6 | | |
| 7 | 9 | | 6 | | 8 | 2 | 5 | |

# Agenti intelligenti

- Esempi tipici di agenti software sono i *bot*
  - In grado di partecipare ad un'asta di Ebay
  - In grado di svolgere automaticamente funzioni necessarie per il sistema operativo (*daemon*)

- Oppure alcune intelligenze artificiali presenti nei videogiochi
  - Dragon Age Origin: ***BDI*** *(Belief-Desires-Intentions)*
  - Sistemi multi-agente (giochi strategici, come Age of Empires)

# AI versus giocatori reali

## 1997

- Deep Blue (IBM) sconfigge Garry Kasparov a scacchi

## 2015

- Alpha Go (Google Deep Mind) sconfigge Fan Hui nel gioco del Go

https://www.deepmind.com/

- Scacchi: *branching factor* > 40, 50 mosse
- Go: *branching factor* > 250, 350 mosse

# Dove può arrivare l’Intelligenza Artificiale?

Riflettiamoci assieme

| AI FORTE | AI DEBOLE |
|---|---|
| Un computer correttamente programmato può essere dotato di intelligenza **non distinguibile da quella umana** | Un computer non sarà **mai in grado di uguagliare la mente umana** |
| Come per il filosofo empirista inglese Thomas Hobbes: **ragionare non è nient’altro che calcolare** | Il ragionamento di un computer non sarà mai così complesso come quello della mente umana |
| Il pensiero umano è il prodotto di un complesso insieme di calcoli eseguiti dal cervello | Un computer potrà solo arrivare a **simulare** alcuni processi cognitivi umani senza riuscire a riprodurli completamente |

What is your perspective on the value of preserving the natural state of other planets and the potential consequences of altering them?

CHATGPT

As an AI, I do not have personal perspective. However, I can provide information and analysis on the topic. One perspective on preserving the natural state of other planets is that it is important to maintain the diversity and integrity of the universe's ecosystems, as they have existed for millions of years, and have their own unique characteristics and properties that may be destroyed or irreversibly altered by human intervention.

Additionally, some argue that terraforming other planets may have unintended consequences, such as altering the planet's climate or atmosphere in ways that are dangerous or harmful to any potential life forms that may exist there. On the other hand, some argue that the potential benefits of terraforming another planet, such as providing new habitable environments for humans, outweigh the risks and that we have a moral responsibility to explore and utilize the resources of other planets for the benefit of humanity.

Would it be moral to terraform Mars?

CHATGPT

The morality of terraforming Mars is a complex and controversial topic that has been debated by scientists, philosophers, and ethicists. Some argue that it would be morally justifiable to terraform Mars in order to provide a new habitable environment for humans and other life forms, while others argue that it would be morally wrong to fundamentally alter the planet's natural state. Ultimately, whether or not terraforming Mars is considered moral would depend on one's perspective on the value of preserving the natural state of other planets and the potential consequences of altering them.

Why can an AI not have a personal perspective?

CHATGPT

As an AI, I am a machine learning model that has been train on large amounts of text data, and I don't have personal experiences or feelings. I am not conscious, self-aware, or capable of having personal perspectives. I can provide information and analysis based on the data I have been trained on, but I don't have the ability to form personal opinions or beliefs. While I can understand and interpret language, I do not have the subjective experience that humans have.

# Ai giorni nostri...

## ChatGPT

https://openai.com/blog/chatgpt

Modello *statistico* di linguaggio

- Imitazione del pensiero umano e del comportamento umano
- Non rientra nell'agire o pensare **razionalmente**!

# Domande?

Lascia la tua opinione sull’orientamento di Informatica!